

Data Warehouses and Data Mining Systems

Report no. 1

Exercise topic:
Defining a problem

Project topic:
Report Flight Delays

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1. Topic

The analysis focuses on flight delays and flight cancellations in U.S. air transport. Using three airline datasets, the report aims to identify when delays occur most often, which airlines and airports are most exposed to disruptions, what the main operational causes are, and how delay patterns change over time. The analysis can support better planning of flight schedules, improvement of airport operations, reduction of cancellation risk, and more accurate communication with passengers.

Dataset Links:

1. https://www.kaggle.com/datasets/patrickzel/flight-delay-and-cancellation-data-2019-2023-v2?utm_source=chatgpt.com

2. https://www.kaggle.com/datasets/bulter22/airline-data/data?utm_source=chatgpt.com

3. <https://www.kaggle.com/datasets/usdot/flight-delays?select=flights.csv>

1.Exemplary business questions:

- Which airlines have the highest average departure and arrival delays?
- Which airports generate the largest number of delayed and cancelled flights?
- At what months, days of the week, and hours do delays occur most frequently?
- What are the main reasons for delays, for example carrier, weather, NAS, security, or late aircraft?
- Which routes are the most vulnerable to cancellations and severe delays?
- How does flight performance differ between years, especially before and after major disruptions in air traffic?
- Which carriers are the most reliable in terms of on-time performance?
- Is departure delay strongly related to arrival delay?
- Which airports act as major bottlenecks in the network?
- Can historical data be used to predict whether a future flight is likely to be delayed or cancelled?

2.Short description of the most important columns

Dataset 1: Flight Delay and Cancellation Data (2019-2023)_v2

- FL_DATE – date of the flight
- AIRLINE – airline name or code operating the flight
- FLIGHT_NUMBER – flight identifier
- ORIGIN – departure airport
- DEST – arrival airport

- CRS_DEP_TIME – scheduled departure time
- DEP_DELAY – departure delay in minutes
- ARR_DELAY – arrival delay in minutes
- CANCELLED – indicates whether the flight was cancelled
- CANCELLATION_CODE – code describing the reason for cancellation
- DIVERTED – indicates whether the flight was diverted

Dataset 2: Airline on-time Performance Data

- Year / Month / DayofMonth / DayOfWeek – calendar variables describing when the flight took place
- UniqueCarrier – airline carrier code
- FlightNum – flight number
- Origin / Dest – origin and destination airports
- CRSDepTime / CRSArrTime – scheduled departure and arrival times
- DepDelay / ArrDelay – departure and arrival delay in minutes
- Cancelled / CancellationCode – information on whether the flight was cancelled and why
- CarrierDelay / WeatherDelay / NASDelay / SecurityDelay / LateAircraftDelay – detailed components of total delay

Dataset 3: 2015 Flight Delays and Cancellations

- flights.csv: YEAR, MONTH, DAY, AIRLINE, FLIGHT_NUMBER, ORIGIN_AIRPORT, DESTINATION_AIRPORT, SCHEDULED_DEPARTURE, DEPARTURE_DELAY, ARRIVAL_DELAY, CANCELLED, DIVERTED, and delay-cause columns
- airlines.csv: IATA_CODE, AIRLINE
- airports.csv: IATA_CODE, AIRPORT, CITY, STATE, COUNTRY, LATITUDE, LONGITUDE

3. What these files contain:

- Dataset 1 contains recent flight-level records from 2019 to 2023. It is useful for studying current delay and cancellation patterns, comparing airlines and airports, and building predictive models on relatively recent data.
- Dataset 2 contains historical U.S. airline on-time performance data from October 1987 to April 2008 and is organized into four tables: flight details, plane information, carrier information, and airport information. It is useful for long-term trend analysis and for explaining delays with richer background data.
- Dataset 3 contains three main CSV files. flights.csv stores detailed information about each 2015 flight, including schedule, delay, cancellation, diversion, and delay causes. airlines.csv contains airline codes and names. airports.csv contains airport identifiers and location-related information.
- Together, these three datasets make it possible to analyze flight delays and cancellations from both a historical and operational perspective, compare airlines and airports, discover seasonal patterns, and prepare data for descriptive analytics, data warehousing, and predictive modeling.